Zitierweise: *Angew. Chem. Int. Ed.* **2023**, *62*, e202303794
doi.org/10.1002/anie.202303794

Nitrogen Reduction Reaction | Hot Paper

Optimizing Electrocatalytic Nitrogen Reduction via Interfacial Electric Field Modulation: Elevating d-Band Center in WS_2-WO_3 for Enhanced Intermediate Adsorption

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Abstract: Electrocatalytic nitrogen reduction reaction (ENRR) has emerged as a promising approach to synthesizing green ammonia under ambient conditions. Tungsten (W) is one of the most effective ENRR catalysts. In this reaction, the protonation of intermediates is the rate-determining step (RDS). Enhancing the adsorption of intermediates is crucial to increase the protonation of intermediates, which can lead to improved catalytic performance. Herein, we constructed a strong interfacial electric field in WS_2-WO_3 to elevate the d-band center of W, thereby strengthening the adsorption of intermediates. Experimental results demonstrated that this approach led to a significantly improved ENRR performance. Specifically, WS_2-WO_3 exhibited a high NH_3 yield of 62.38 $\mu g\, h^{-1} mg_{cat}^{-1}$ and a promoted faraday efficiency (FE) of 24.24 %. Furthermore, in situ characterizations and theoretical calculations showed that the strong interfacial electric field in WS_2-WO_3 upshifted the d-band center of W towards the Fermi level, leading to enhanced adsorption of $-NH_2$ and $-NH$ intermediates on the catalyst surface. This resulted in a significantly promoted reaction rate of the RDS. Overall, our study offers new insights into the relationship between interfacial electric field and d-band center and provides a promising strategy to enhance the intermediates adsorption during the ENRR process.

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Introduction

Ammonia (NH_3) is an indispensable chemical in agriculture, chemical industry, refrigeration, hydrogen energy and healthcare.[1–3] However, the conventional Haber-Bosch process, which dominates industrial ammonia synthesis, requires high temperature and pressure, resulting in a significant environmental impact.[4–6] To address this issue, electrocatalytic nitrogen reduction reaction (ENRR) is being considered as an alternative approach to produce NH_3 under mild conditions, which is more sustainable and environmentally friendly.[7–9] However, the sluggish reaction kinetics significantly limit the activity and selectivity of ENRR.[10–13] Hence, it is highly urgent to develop an efficient catalyst to promote the ENRR performance.

The electrochemical reduction of N_2 on the surface of catalysts involves two steps: 1) the adsorption and activation of N_2, and 2) the subsequent protonation of adsorbed intermediates.[14–16] According to recent studies, the rate-determining step (RDS) for early transition metals (Sc to Fe and Y to Rh) is the protonation of intermediates, whereas for later transition metals (Co, Ni, and Pt/Pd), it is the adsorption and activation of N_2. Among these catalysts, W-based catalysts are particularly effective for ENRR due to its intrinsically weak binding with hydrogen.[17,18] Improving the adsorption of intermediate products generated during ENRR can significantly accelerate the reaction rate of the RDS of W-based catalysts and improve their overall performances. Elevating the d-band center of W toward the Fermi level is an effective method for strengthening the adsorption of intermediates. According to ligand field theory, the energy levels of the d orbitals of metal atoms are influenced by the surrounding ligands, which can donate or accept electrons from the metal atom. The incorporation of heteroatoms into the metal lattice modifies the electronic structure of the metal, resulting in changes to the energy levels of the metal atoms' d orbitals and consequently altering the center of the d-band. For example, Sun et al. incorporated Cu into carbon nitride to upshift the d-band center of Cu, resulting in enhanced adsorption and activation of CO_2 and thus facilitating efficient conversion of CO_2 to CO.[19] Li et al. introduced Ni into CoS_2 to enhance the adsorption capacities of intermediates by elevating the d-band center, resulting in improved kinetics of both the oxygen reduction reaction and the oxygen evolution reaction.[20] While numerous efforts have been made to shift the d-band center upwards, there have been relatively few studies investigating the tuning of the d-band center through constructing an interfacial electric field in catalysts.

In this study, we have successfully constructed a strong interfacial electric field in a WS_2-WO_3 heterostructure to enhance W, resulting in accelerated ENRR kinetics. The existence of heterogeneous interfaces in WS_2-WO_3 was confirmed annular dark-field scanning transmission electron microscopy (HAADF-STEM). Kelvin probe force microscopy (KPFM) and zeta potential measurements provided evidence for a strong interfacial electric field formed in WS_2-WO_3. Theoretical calculations affirmed that a stronger interfacial electric field in WS_2-WO_3 could upshift the d-band center of W, which plays a vital role in the adsorption of intermediates. Furthermore, in situ Raman and Fourier transform infrared (FTIR) spectra offered direct evidence for enhanced adsorption of $*NH_2$ and $*NH$ intermediates. These findings provide deep insights into the effects of interfacial electric field on the ENRR catalysts performance and offer a promising approach for the development of sustainable and efficient catalysts in other related fields.

Results and Discussion

To prove the rationality of the d-band center model toward W based catalysts for ENRR, density functional theory (DFT) calculations were employed to predict the effects of the construction of interfacial electric field on the d-band center. We generally categorize the interfaces between WS_2 and WO_3 into three cases prior to optimization, as depicted in Figure S1. The results show that the model 2 is the most stable interface with low energy, which indicates that it is the easiest obtained structure in experiment. The different views model WS_2, WO_3 and WS_2-WO_3 are displayed in Figure S2. In addition, the electrostatic potential difference between the top and bottom of the WS_2-WO_3 heterostructure was calculated under various external electric fields (EF), ranging from 0.1 V to 0.5 V Å^{-1}, as shown in Table S1. The results demonstrate that the EF can effectively change the interfacial electric field within the WS_2-WO_3 system. The projected density of states (PDOS) of WS_2-WO_3 showed that a stronger interfacial electric field can elevate the d-band centers (ε_d) of W (Figure 1a), and a roughly negative linear relationship between the ε_d value and interfacial electric field intensity was observed (Figure 1b).[21] Besides, the projected Crystal Orbital Hamilton Population (pCOHP) was conducted for WS_2-WO_3 with different interfacial electric field intensity (Figure S3). The integrated Crystal Orbital Hamilton Population (ICOHP) was calculated by integrating all states up to Fermi level.[22,23] A positive linear relationship between ICOHP and interfacial electric field intensity was confirmed (Figure 1c) with results suggesting that the occupancy of the antibonding states decreases with increased interfacial electric field. In a manner analogous to ligand field theory, the interfacial electric field can promote electron rearrangement within the catalyst and modulate the electronic structure of the metal. Thus, a strong interfacial electric field could obviously

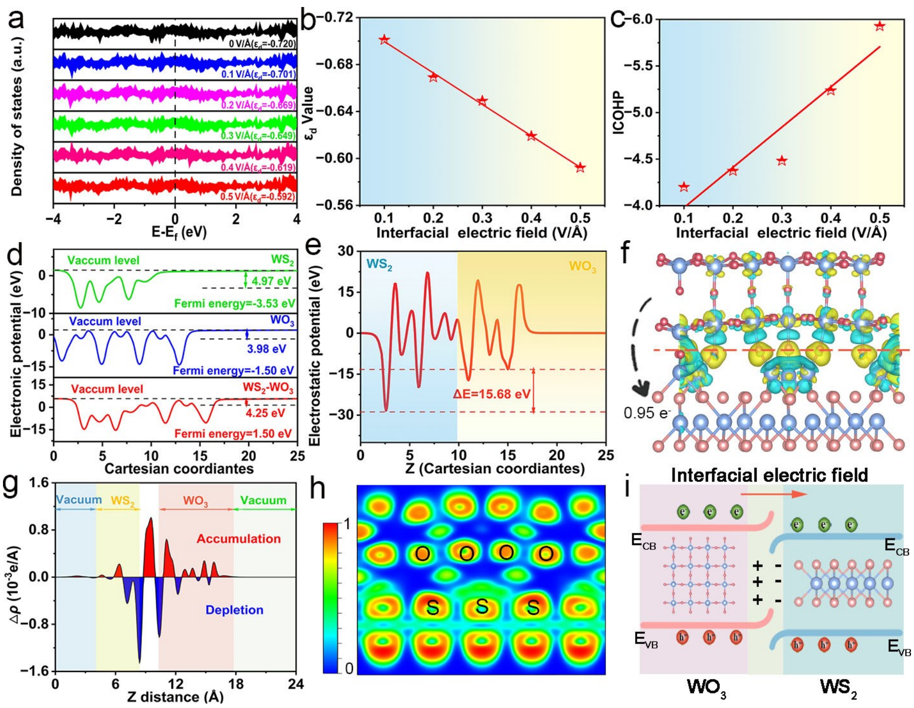

Figure 1. a) The PDOS of WS_2-WO_3 with different interfacial electric field intensities and d-band center (ε_d) of the active W sites. b) Relationship between ε_d values and interfacial electric field intensities in WS_2-WO_3. c) Relationship between ICOHP values and interfacial electric field intensities in WS_2-WO_3. d) The calculated WF of WS_2, WO_3 and WS_2-WO_3. e) Electrostatic potential profile of WS_2-WO_3. f) The DCD of WS_2-WO_3. g) Plane-averaged charge density difference for WS_2-WO_3 along z-direction. h) ELF of WS_2-WO_3. i) Schematic illustration of the transfer process of interface charge in WS_2-WO_3 heterostructure.

upshift the d-band center and lower the occupancy of antibonding states of W, which should be beneficial to the adsorption of the intermediates generated during ENRR. To understand the formation mechanism of interfacial electric field in WS_2-WO_3, a series of calculations were performed to evaluate the electron transfer processes between WS_2 and WO_3. Firstly, the work functions (WF) of WS_2, WO_3 and WS_2-WO_3 were calculated to be 4.97, 3.98 and 4.25 eV (Figure 1d), respectively. A smaller WF value of WO_3 implies that its electrons can more easily escape from the surface compared to WS_2.[24,25] This suggests that electrons are more likely to transfer from WO_3 to WS_2 at the WS_2-WO_3 interface. Besides, an electron transfer potential ($\Delta V_{et} = 15.68$ eV) for WS_2-WO_3 was obtained between WS_2 region and WO_3 region (Figure 1e), indicating that the electrons are more likely to transfer from WO_3 to WS_2. Secondly, the electronic structure of WS_2-WO_3 was investigated through the difference charge density analysis and electronic local function (ELF). As displayed in Figure 1f, electronic transfer of $0.95\,e^-$ from WO_3 to WS_2 at the interface of WS_2-WO_3 was observed. Besides, the z-direction plane-averaged charge density difference of WS_2-WO_3 was shown in Figure 1g, in which the red and blue regions are electron accumulation and depletion, respectively. Obviously, WS_2-WO_3 heterostructure model exhibited abundant charge transfer from WO_3 to WS_2, resulting in forming a strong interfacial electrical field.[26–28] Furthermore, the region of WS_2 close to 1 has a higher electron density compared to the region of WO_3 (Figure 1h), further suggesting electrons have transferred from WO_3 to WS_2. Figure 1i shows the process of charge transfer in WS_2-WO_3 heterostructure to vividly present the formation of interfacial electric field. Overall, the above calculations suggest that a strong interfacial electric field could be constructed in the WS_2-WO_3 heterostructure to elevate the d-band center of W, thereby likely strengthening the adsorption of intermediates and facilitating the ENRR kinetics.

To confirm the above hypothesis, the WS_2-WO_3 heterostructure was constructed using an annealing treatment method, as schematically illustrated in Figure 2a. The X-ray diffraction (XRD) results (Figure S4) of WS_2-WO_3 exhibited peaks at 14.3°, 28.9°, 43.9° and 60.5°, ascribing to the (002), (004), (006) and (112) typical diffraction planes of the WS_2 (JCPDS No. 08-0237), and peaks at 23.1°, 24.4°, 26.6° and 34.2°, attributing to the (002), (200), (202) and (120) planes of the WO_3 (JCPDS No. 43-1035), respectively, manifesting the successful synthesis of WS_2-WO_3. Furthermore, we use scan-

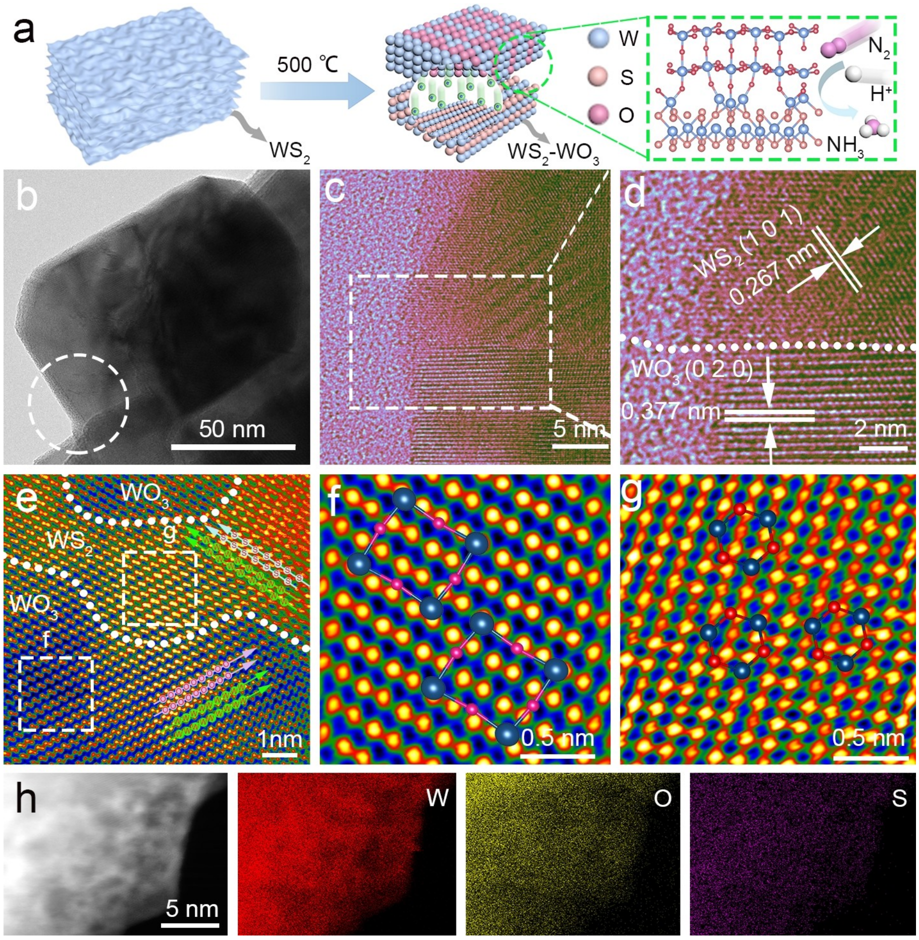

Figure 2. a) Schematic diagram of WS_2-WO_3 synthesis. b) TEM image of WS_2-WO_3; c) HRTEM image of the selected region in (b). d) Enlarged view of the area indicated by the white rectangular in (c). e) HAADF-STEM image of WS_2-WO_3. f) FFT patterns of the area f indicated by the white rectangular in (e). g) FFT patterns of the area g indicated by the white rectangular in (e). h) HAADF mapping images of WS_2-WO_3.

ning electron microscopy (SEM) and transmission electron microscope (TEM) to elucidate the morphology of all samples. Figure S5 and Figure 2b showed that WS_2-WO_3 was composed of stacked sheets, and pure WS_2 (Figures S6a, b) and WO_3 samples (Figures S7a, b) were also composed of stacked sheets. The high-resolution TEM (HRTEM) images showed that the lattice spacing of 0.267 nm and 0.377 nm corresponded to the WS_2 (101) planes and WO_3 (020) planes (Figures 2c, d), respectively. Clearly, the abundant contacting interfaces between WS_2 and WO_3 components were observed in HRTEM images. Simultaneously, HRTEM images of WS_2 (0.309 nm) and WO_3 (0.524 nm) samples were displayed in Figure S6c and Figure S7c. To precisely investigate the structure of WS_2-WO_3, the HAADF-STEM was further employed. As displayed in Figure 2e, the atoms at the interface of WS_2 and WO_3 were shown as light green, blue, and pink spheres to represent the W, S, and O atoms, respectively. Furthermore, fast Fourier transform (FFT) patterns of the area marked with a white rectangular in Figure 2e were acquired. WS_2 sheets were hexagonal crystal (Figure 2f), and WO_3 sheets were orthorhombic crystal system (Figure 2g). Notably, the observed interfaces between WS_2 and WO_3 components in WS_2-WO_3 indicated the generation of heterostructure with abundant interface contact. As shown in Figure 2h, the elemental mapping revealed that the W, O, and S elements were uniformly distributed in the WS_2-WO_3. As a contrast, the mapping images of WS_2 and WO_3 were shown in Figure S6d and Figure S7d. In general, these results definitively demonstrated the WS_2-WO_3 with an obvious interface between WS_2 and WO_3 components was successfully preparate. Furthermore, X-ray photoelectron spectroscopy (XPS) was used to study the chemical and valence states of all samples. According to Figure S8a, the O 1s spectrum of WS_2-WO_3 exhibited a peak at 533.73 and 531.7 eV, according with adsorbed and lattice O species, respectively. These peaks were positive shifted by 1.44 eV and 1.43 eV compared to WO_3 (532.29 and 530.27 eV), respectively.[29] The S 2p spectrum showed a pair of peaks at 164.01 eV and 165.27 eV, which were assigned to S $2p_{3/2}$ and S $2p_{1/2}$, respectively (Figure S8b). These peaks were negative shifted by 0.61 eV and 0.53 eV compared to WS_2 (164.62 and 165.80 eV), respectively.[30] The XPS peak shift evidently proved electron transfer from WO_3 to the WS_2 sheets, indicating the strong interactions between WS_2 and WO_3 at the interface. As shown in Figure S8c, the peaks for WS_2-WO_3 at 33.46, 35.63 and 38.67 eV attributed to W^{4+} of 4 $f_{7/2}$, 4 $f_{5/2}$ and 5 $p_{5/2}$, respectively, and other peaks at 36.45 and 38.02 eV ascribed to the W^{6+}4 $f_{7/2}$ and W^{6+}4 $f_{5/2}$, respectively.[31]

UV/Vis diffuse reflectance spectra (DRS) and Mott–Schottky curves were carried out to investigate the band structure of the samples. The Mott–Schottky curves indicated that both WS_2 and WO_3 exhibited n-type semiconductor traits with a positive slope. The flat band potential (E_{fb}) of WS_2 and WO_3 (Figure 3a) were −0.72 V and −0.81 V (vs. Ag/AgCl), respectively. Typically, the conduction band (CB) potential of n-type semiconductors is 0.1 V lower than E_{fb}, which indicated that the CB positions of WS_2 and WO_3 were −0.82 V and −0.91 V, respectively.[32,33] The band gap (E_g) of WS_2 and WO_3 were determined to be 1.37 and 2.58 eV, respectively, via Kubelka–Munk equation (Figure 3b).[32,34] A schematic for the band structures of WS_2 and WO_3 was presented in Figure 3c, which illustrated that electrons transferred from WO_3 to WS_2, in agreement with the XPS results. Furthermore, ultraviolet photoelectron spectroscopy (UPS) analysis was performed to investigate the interactions between WS_2 and WO_3 phases in the WS_2-WO_3 heterostructure. The work functions of WS_2, WO_3, and WS_2-WO_3 were calculated to be 10.37, 9.25, and 9.98 eV (Figure 3d), respectively, which was in agreement with DFT calculation results. WO_3 had a lower work function than WS_2, indicating that electrons could transfer from WO_3 to WS_2.[35,36] To obtain the surface charge, the zeta potential measurement was performed. As displayed in the Figure 3e, the acquired zeta potential of WS_2-WO_3 was −35.1 mV, which was 1.2 times and 1.8 times higher than that of WS_2 (−29.5 mV) and WO_3 (−19.2 mV), respectively.[37,38] Furthermore, KPFM was utilized to measure the surface potential of catalysts.[34,39,40] Three-dimensional (3D) surface potential distribution images of WS_2, WO_3, and WS_2-WO_3 were obtained, respectively, as presented in Figures 3g–i. The observed corresponding line-scanning surface potential profiles demonstrated that the contact potential difference (CPD) of WS_2-WO_3 was 35.31 mV, which was much higher than that of WS_2 (17.29 mV) and WO_3 (9.49 mV). This demonstrated that WS_2-WO_3 possessed a stronger interfacial electric field. Besides, the intensity of interfacial electric field was quantitatively characterized by differentiating the fitting surface potential profiles (Figures 3f and S9). The interfacial electric field intensity of WS_2-WO_3 delivered 0.067 V μm^{-1}, which was maximum among all samples (WS_2 0.034 V μm^{-1} and WO_3 0.019 V μm^{-1}). These results convincingly affirmed that the WS_2-WO_3 heterostructure formed a strong interfacial electric field between the WS_2 and WO_3 phases, which was anticipated to be beneficial for improving the performance of WS_2-WO_3 for ENRR.

The ENRR performances of all the samples were measured in 0.1 M Li_2SO_4 solutions using an H-type cell under ambient conditions (Figure S10). Compared to Ar-saturated electrolyte, the linear sweep voltammogram (LSV) curves of WS_2-WO_3 (Figure 4a) in N_2-saturated electrolyte showed an increased reduction current density, indicating the occurrence of ENRR on WS_2-WO_3. Meanwhie, the step voltage chronoamperometry experiment recorded the real ENRR currents for WS_2-WO_3 at −0.4 V (vs. RHE), which is higher than that of the open circuit potential (Figure 4b). Moreover, the obtained currents in N_2 were higher compared to Ar atmosphere. The above results demonstrated that the WS_2-WO_3 surface was essentially proceeded ENRR at the given potential. Furthermore, chronoamperometry tests and the UV/Vis absorption spectra after NRR for 2 h from −0.2 to −0.6 V versus RHE were collected to determine the amount of ammonia (Figure S13 and Figure S14). To facilitate comparison, we also prepared a physically mixed WS_2 and WO_3 sample (referred to as WS_2&WO_3) for electrochemical studies. The NH_3 yield was quantified by the indophenol-blue method (Figure S11) and the byproduct N_2H_4 was measured by Watt and Chrisp method (Figure S12). No N_2H_4 was produced during the ENRR process (Figure 4c), indicating the high ENRR selectivity of WS_2-WO_3. Figure 4d showed the NH_3 yields obtained at different applied potentials using WS_2, WO_3,

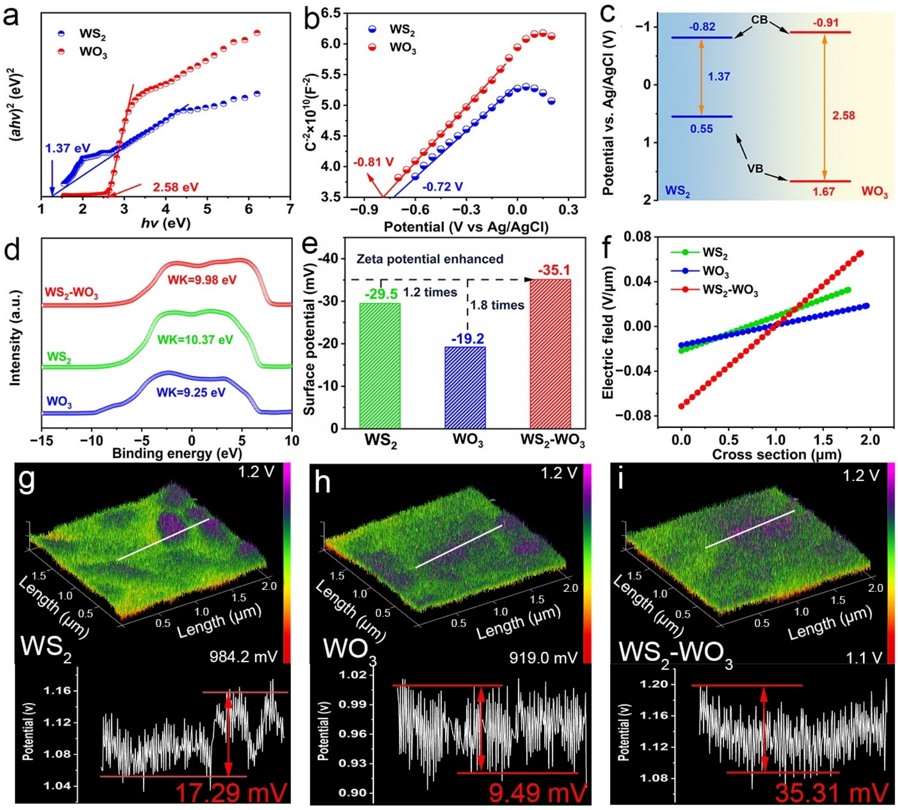

Figure 3. a) Mott–Schottky plots and b) Kubelka–Munk plots of WS_2 and WO_3. c) Electronic band structure of WS_2 and WO_3. d) Secondary electron cutoff edge of WS_2, WO_3 and WS_2-WO_3 acquired by UPS. e) The zeta potentials of WS_2, WO_3 and WS_2-WO_3. f) The interfacial electric field intense distributions on WS_2, WO_3 and WS_2-WO_3 surfaces. g)–i) 3D surface potential distributions and corresponding lines canning surface potential profiles of WS_2, WO_3 and WS_2-WO_3.

WS_2&WO_3 and WS_2-WO_3 as the catalysts, respectively. The obtained maximum NH_3 yield value of WS_2-WO_3 was 62.38 $\mu g\,h^{-1}\,mg_{cat}^{-1}$ at -0.40 V (Figure 4d), which was obvious higher than that of WS_2 (40.41 $\mu g\,h^{-1}\,mg_{cat}^{-1}$), WO_3 (36.55 $\mu g\,h^{-1}\,mg_{cat}^{-1}$) and WS_2&WO_3 (36.82 $\mu g\,h^{-1}\,mg_{cat}^{-1}$). Additionally, the highest FE of WS_2-WO_3 was 24.24 % (Figure 4e), which was also higher than that of WS_2 (13.99 %), WO_3 (13.01 %) and WS_2&WO_3 (13.46 %). In comparison, the obtained performance of WS_2-WO_3 was higher than that of most recently reported ENRR electrocatalysts (Table S3). Simultaneously, ion chromatograph (IC) was employed to double examine NH_3 production (Figure S15). The yield of ammonia for the WS_2-WO_3 under optimized condition was calculated to be 61.03 $\mu g\,h^{-1}\,mg_{cat}^{-1}$, which was in close consistent with the colorimetric result (62.38 $\mu g\,h^{-1}\,mg_{cat}^{-1}$). Thus, the IC results certified the reliability of the results by spectrophotometry. Furthermore, the electrochemical stability of WS_2-WO_3 was evaluated at -0.4 V versus RHE (Figure S16) with results showing that NH_3 yields rate and FEs of WS_2-WO_3 were maintained consistently over five consecutive cycles (each cycle lasted two hours), implying the excellent electrochemical stability of WS_2-WO_3. Besides, the diffraction peaks of XRD for WS_2-WO_3 showed no obvious changes after the long-time test in Figure S17.

To prove the N source of the produced NH_3 during ENRR, a series of controlled experiments were carried out. The results of these experiments, as depicted in Figure S18, showed that no NH_3 products were detected in all control conditions. To further investigate the origin of the produced NH_3, isotopic labeling experiments were conducted using $^{15}N_2$ (99 atom% ^{15}N) and $^{14}N_2$ as the feeding gas. The ^1H nuclear magnetic resonance (^1H NMR) spectra revealed double coupling and triplet coupling for $^{15}NH_4^+$ and $^{14}NH_4^+$ (Figure 4f), respectively, indicating that the generated NH_3 came from ENRR.[41] Moreover, the double-layer capacitances of the samples were calculated based on the CV curves and used to evaluate the electrochemically active surface area (ECSA) of each sample (Figure S19).[41,42] As illustrated in Figure 4g, the capacitance of WS_2-WO_3 (1.621 $\mu F\,cm^{-2}$) was smaller than that of WS_2 (2.576 $\mu F\,cm^{-2}$) and WO_3 (1.824 $\mu F\,cm^{-2}$), indicating that WS_2-WO_3 possessed a smaller ECSA with fewer active sites. Nonetheless, the NH_3 yield of WS_2-WO_3 electrocatalyst outperformed that of the other samples, suggesting that the ECSA is not a crucial factor for ENRR, and the enhanced ENRR kinetics on WS_2-WO_3 could be ascribed to the interfacial electric field. Moreover, electrochemical impedance spectroscopy (EIS) tests were conducted to illustrate the charge transfer behavior (Figure 4h). The WS_2-WO_3 electrocatalyst possessed the lowest charge transfer resistance (R_{ct}) value among all samples, indicating better charge transfer kinetics due to the engineered interfacial electric field.

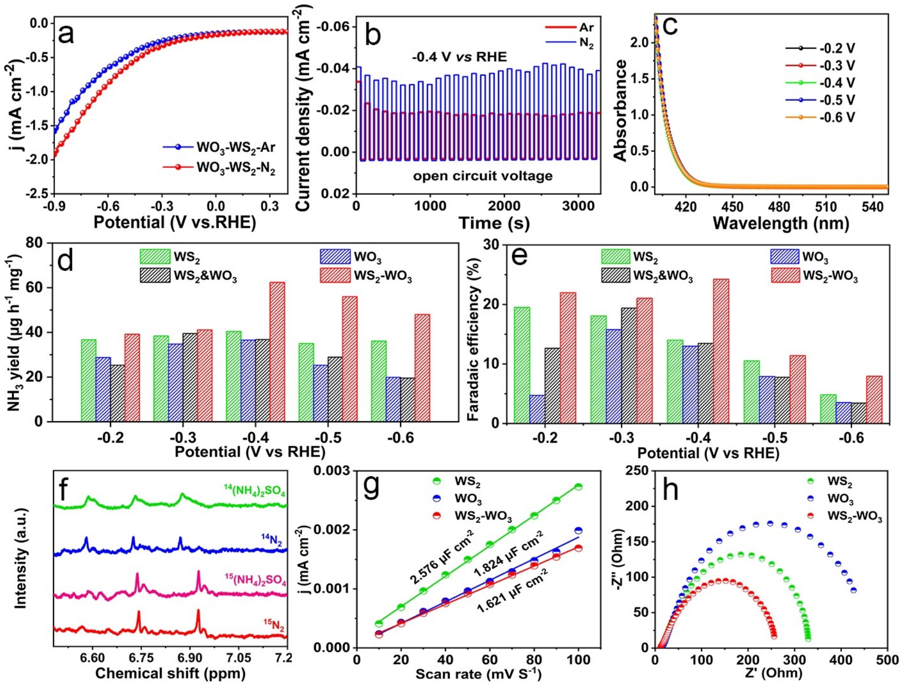

Figure 4. a) LSV curves of WS_2-WO_3 in 0.1 M Li_2SO_4 under Ar-saturated and N_2-saturated conditions. b) Voltage chronoamperometry curves of WS_2-WO_3 at −0.4 V vs RHE and open-circuit potential. c) UV/Vis absorption spectra of the electrolytes with the Watt and Chrisp's method. d) NH_3 yields of WS_2, WO_3, WS_2&WO_3 and WS_2-WO_3. e) Faradaic efficiencies of WS_2, WO_3, WS_2&WO_3 and WS_2-WO_3. f) ^1H NMR spectra of WS_2-WO_3 fed by $^{14}N_2$ and $^{15}N_2$ gas. g) C_{dl} measurement of WS_2, WO_3 and WS_2-WO_3 0.1 M Li_2SO_4 electrolyte. h) Nyquist plots of WS_2, WO_3 and WS_2-WO_3 in 0.1 M Li_2SO_4 electrolyte.

To understand the underlying mechanism for the enhanced performance of WS_2-WO_3 for ENRR, the PDOS, in situ Raman and in situ FTIR spectra were measured to describe the binding strength between catalysts and intermediates. The d-band center position of the catalyst is an important factor that determines the adsorption energy of intermediates.[43] An upshifted d-band center towards the Fermi level indicates enhanced adsorption of intermediates. This is because the higher energy level of the d-band center allows for stronger interaction between the catalyst and intermediates, leading to more efficient catalytic reactions. WS_2-WO_3 has a significantly upshifted d-band center (−0.72 eV) compared to WS_2 (−1.57 eV) and WO_3 (−3.24 eV) (Figure 5a), indicating that WS_2-WO_3 should possess a stronger binding strength for ENRR intermediates.[44,45] Furthermore, in situ Raman spectroscopy was used to monitor the possible reaction intermediates during potentiation operation for 1 h. Two peaks at approximately 1328 and 1574 cm^{-1} were observed, which were assigned to $-NH_2$ and –NH, respectively (Figures 5b and S20).[16,46] The adsorption intensities of $-NH_2$ and –NH on the surface of WS_2-WO_3 were higher compared to those on the surfaces of WS_2 and WO_3, indicating that the surface of WS_2-WO_3 has a stronger binding strength for ENRR intermediates. Moreover, five positive peaks located at 1148, 1209, 1504, 1536 and 3452 cm^{-1} were observed, which could be attributed to N–N stretching, $-NH_2$ wagging, –H–N–H bending, $-NH_4^+$ wagging, and –N–H bending, respectively (Figures 5c and S21).[12,46] The intensities for the $-NH_2$ and –NH adsorption on the surface of WS_2-WO_3 were also stronger than those on the surfaces of WS_2 and WO_3. Above findings demonstrate that the upshifted d-band center in WS_2-WO_3 can effectively enhance the adsorption of $-NH_2$ and –NH intermediates, which is responsible for its superior electrocatalytic activity towards ENRR.

To gain a deeper understanding of the mechanism underlying the enhanced intermediate adsorption capacity on the surface of WS_2-WO_3, the DFT calculations were employed. The ICOHP values of WS_2, WO_3, and WS_2-WO_3 were calculated by integrating all states up to the Fermi level and found to be −3.18, −1.50, and −4.08 (Figures 5d–f), respectively. The results indicate that the antibonding state energy level of WS_2-WO_3 is lower than those of WS_2 and WO_3, leading to increased intermediate adsorption. Additionally, the adsorption energies of $*NH_2$ and *NH intermediates were also determined to illustrate the adsorption capability of WS_2-WO_3. Figure 5g confirms that WS_2-WO_3 has more negative adsorption energies for $*NH_2$ and *NH (−1.45 and −1.34 eV) than WS_2 (−0.29 and −1.01 eV) and WO_3 (−0.18 and −0.73 eV), indicating stronger binding of $*NH_2$ and *NH on the WS_2-WO_3 surface. Furthermore, the projected density of states (PDOS) of WS_2, WO_3, and WS_2-WO_3 upon adsorption of $*NH_2$ and *NH were analysed.

The position of E_p (highest peak of active centre below the Fermi level) is an excellent descriptor to investigate the binding strength between intermediates and electrocatalysts.[47] A higher E_p position means that the antibonding states of the intermediates are located higher, with lower occupancy, which results in stronger binding of the intermediates on the electro-

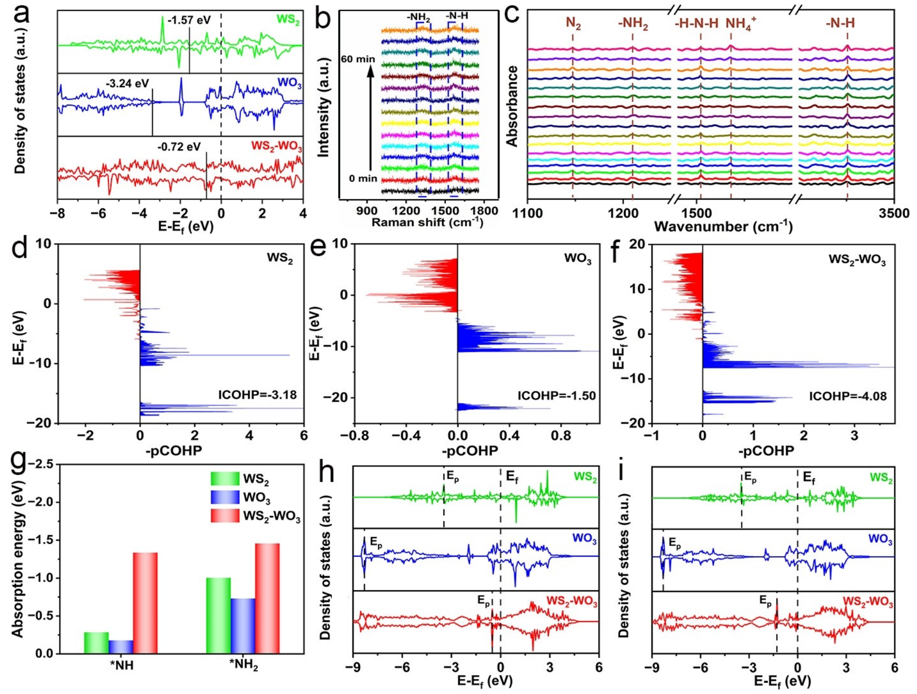

Figure 5. a) The PDOS of WS_2, WO_3 and WS_2-WO_3 for the active W sites, respectively. b) In situ Raman spectra of WS_2-WO_3 at −0.4 V in 0.1 M Li_2SO_4. c) In situ FTIR spectra of WS_2-WO_3. The pCOHP of (d–f) WS_2, WO_3 and WS_2-WO_3. g) Calculated adsorption energies of active intermediates. DOS of the h) *NH and i) *NH_2 intermediates on WS_2, WO_3 and WS_2-WO_3.

catalyst surface. As shown in Figures 5h and i, the E_p position of WS_2-WO_3 is higher than that of WS_2 and WO_3 upon the adsorption of *NH_2 and *NH, indicating the stronger adsorption of *NH_2 and *NH on the surface of WS_2-WO_3. Additionally, when comparing the PDOS before adsorption of *NH_2 and *NH, it is evident that the spin splitting changes. Consequently, the magnetism undergoes alterations during the reaction process, which can be attributed to the varying charge transfer from the W site to the adsorbates throughout the reaction process (Table S2). To evaluate the catalytic activity of WS_2-WO_3, ENRR mechanism on the surface of WS_2-WO_3 was investigated in detail. We calculated the free energy diagram of WS_2-WO_3 under its inherent interfacial electric field. For comparison, the free energy diagrams of WS_2 and WO_3 were calculated without electric field. Figures S22–S24 present a comparison of the ENRR free-energy profiles of the WS_2, WO_3 and WS_2-WO_3 following the distal mechanism. Interestingly, the RDS for WS_2 and WO_3 is formation *NNH, and its ΔG is 2.388 eV and 2.593 eV, respectively. The RDS for WS_2-WO_3 is formation *N, and its ΔG is 1.806 eV, which indicates that the WS_2-WO_3 optimizes the free-energy barrier of the RDS. The computational results demonstrate that the interfacial electric field in WS_2-WO_3 heterostructure can accelerate the interface electron transfer and regulate the electronic structure of active sites, which effectively changes the RDS from *NNH formation to *N formation with a reduced energy barrier.

Conclusion

In summary, inspired by the theoretical calculation results that the construction of strong interfacial electric field can elevate the d-band center towards Fermi level, we developed the WS_2-WO_3 with a strong interfacial electric field as a high-efficiency catalyst for ENRR. In situ characterizations revealed that the upshifted d-band center of W in WS_2-WO_3 can enhance the adsorption of –NH_2 and –NH intermediates, resulting in a high NH_3 yield of 62.38 $\mu g\,h^{-1}\,mg_{cat}^{-1}$ and a promoted FE of 24.24 % at −0.4 V vs RHE. Furthermore, DFT calculations affirmed that WS_2-WO_3 possessed lower energy barriers for –NH_2 and –NH adsorption, and thus the RDS were changed from *NNH formation to *N formation with a reduced energy barrier on the surface of WS_2-WO_3. This study reveals that the interfacial electric field in WS_2-WO_3 can obviously facilitate ENRR by tuning the d-band center of W and the adsorption of intermediates, which provides a guideline for the rational design of catalysts for ENRR and other related reactions.

Acknowledgements

Financial support from the National Natural Science Foundation of China (grant nos. 21575016, U20A20154 and 22279005) and from the National Program for Support of Top-notch Young Professionals is gratefully acknowledged.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the Supporting Information of this article.

Keywords: d-Band Center and Intermediates Adsorption · Electrochemical Nitrogen Reduction · Interfacial Electric Field

[1] Z. Fang, P. Wu, Y. M. Qian, G. H. Yu, *Angew. Chem. Int. Ed.* **2021**, *60*, 4275–4281.
[2] L. Wen, K. Sun, X. Liu, W. Yang, L. Y. Li, H. L. Jiang, *Adv. Mater.* **2023**, *35*, 2210669.
[3] Y. Yang, L. Zhang, Z. Hu, Y. Zheng, C. Tang, P. Chen, R. Wang, K. Qiu, J. Mao, T. Ling, S. Z. Qiao, *Angew. Chem. Int. Ed.* **2020**, *59*, 4525–4531.
[4] Y. Zhao, F. S. Li, W. Li, Y. Li, C. Liu, Z. Zhao, Y. Shan, Y. F. Ji, L. C. Sun, *Angew. Chem. Int. Ed.* **2021**, *60*, 20331–20341.
[5] J. Zhou, X. Chen, M. Guo, W. Hu, B. W. Huang, D. W. Yuan, *ACS Catal.* **2023**, *13*, 2190–2201.
[6] X. Lin, Y. Wang, X. Chang, S. Zhen, Z. J. Zhao, J. L. Gong, *Angew. Chem. Int. Ed.* **2023**, *62*, e202300122.
[7] K. Ba, G. Wang, T. Ye, X. Wang, Y. Sun, H. Liu, A. Hu, Z. Y. Li, Z. Z. Sun, *ACS Catal.* **2020**, *10*, 7864–7870.
[8] G. Lin, Q. Ju, X. Guo, W. Zhao, S. Adimi, J. Ye, Q. Bi, J. Wang, M. H. Yang, F. Q. Huang, *Adv. Mater.* **2021**, *33*, 2007509.
[9] M. Kim, J. Chung, T. An, J. Lee, M. Han, S. Lee, W. Choi, J. Kim, S. Han, U. Sim, T. Y. Yu, *Appl. Catal. B* **2023**, *328*, 122485.
[10] J. Kang, X. Chen, R. Si, X. Gao, S. Zhang, G. Teobaldi, A. Selloni, M. Liu, L. Guo, *Angew. Chem. Int. Ed.* **2023**, *62*, e202217428.
[11] S. Qiang, F. Wu, J. Yu, Y. Liu, B. Ding, *Angew. Chem. Int. Ed.* **2023**, *62*, e202217265.
[12] Y. Li, Y. Ji, Y. Zhao, J. Chen, S. Zheng, X. Sang, B. Yang, Z. Li, L. Lei, Z. Wen, X. L. Feng, Y. Hou, *Adv. Mater.* **2022**, *34*, 2202240.
[13] M. Han, M. Guo, Y. Yun, Y. Xu, H. Sheng, Y. Chen, Y. Du, K. Ni, Y. Zhu, M. Z. Zhu, *Adv. Funct. Mater.* **2022**, *32*, 2202820.
[14] Y. Kong, Y. Li, X. Sang, B. Yang, Z. Li, S. Zheng, Q. Zhang, S. Yao, X. Yang, L. C. Lei, S. Zhou, G. Wu, Y. Hou, *Adv. Mater.* **2022**, *34*, 2103548.
[15] L. Shi, Y. Yin, S. Wang, H. Q. Sun, *ACS Catal.* **2020**, *10*, 6870–6899.
[16] D. Chen, M. Luo, S. Ning, J. Lan, W. Peng, Y. R. Lu, T. Chan, Y. W. Tan, *Small* **2022**, *18*, 2104043.
[17] B. H. R. Suryanto, H. L. Du, D. Wang, J. Chen, A. N. Simonov, D. R. MacFarlane, *Nat. Catal.* **2019**, *2*, 290–296.
[18] Y. Tong, H. Guo, D. Liu, X. Yan, P. Su, J. Liang, S. Zhou, J. Liu, G. Q. Lu, S. X. Dou, *Angew. Chem. Int. Ed.* **2020**, *59*, 7356–7361.
[19] X. Sun, L. Sun, G. Li, Y. Tuo, C. Ye, J. Yang, J. Low, X. Yu, J. Bitter, Y. Lei, D. S. Wang, Y. D. Li, *Angew. Chem. Int. Ed.* **2022**, *61*, e202207677.
[20] D. Y. Li, L. Zhao, J. Wang, C. P. Yang, *Adv. Energy Mater.* **2023**, *13*, 2204057.
[21] Z. Jiang, S. Song, X. Zheng, X. Liang, Z. X. Li, H. Gu, Z. Li, Y. Wang, S. H. Liu, W. Chen, D. Wang, Y. D. Li, *J. Am. Chem. Soc.* **2022**, *144*, 19619–19626.
[22] F. Xiao, Y. Wang, G. Xu, F. Yang, S. Zhu, C. Sun, Y. Cui, Z. Xu, Q. Zhao, J. Jang, X. Qiu, E. Liu, W. S. Drisdell, Z. D. Wei, M. Gu, K. L. Amine, M. H. Shao, *J. Am. Chem. Soc.* **2022**, *144*, 20372–20384.
[23] Y. Gu, B. Xi, W. Tian, H. Zhang, Q. Fu, S. L. Xiong, *Adv. Mater.* **2021**, *33*, 2100429.
[24] Y. Zhou, Y. B. Yao, R. Zhao, X. Wang, Z. Z. Fu, D. Wang, H. Wang, L. Zhao, W. Ni, Z. Y. Yang, Y. M. Yan, *Angew. Chem. Int. Ed.* **2022**, *61*, e202205832.
[25] L. Lv, X. He, J. Wang, Y. Ruan, S. Ouyang, H. Yuan, T. R. Zhang, *Appl. Catal. B* **2021**, *298*, 120531.
[26] M. Yuan, J. Chen, Y. Bai, Z. Liu, J. Zhang, T. Zhao, Q. Wang, S. Li, H. He, G. Zhang, *Angew. Chem. Int. Ed.* **2021**, *60*, 10910–10918.
[27] Z. Zhu, Q. Lv, Y. Ni, S. Gao, J. Geng, J. Liang, F. Li, *Angew. Chem. Int. Ed.* **2022**, *61*, e202116699.
[28] L. Xu, B. Tian, T. Wang, Y. Yu, Y. Wu, J. Cui, Z. Cao, J. Wu, W. Zhang, Q. Zhang, J. Liu, Z. Li, Y. Tian, *Energy Environ. Sci.* **2022**, *15*, 5059–5068.
[29] P. Zhou, Q. Xu, H. Li, Y. Wang, B. Yan, Y. Zhou, J. Chen, J. Zhang, K. Wang, *Angew. Chem. Int. Ed.* **2015**, *54*, 15226–15230.
[30] L. Yang, X. Zhu, S. Xiong, X. Wu, Y. Shan, P. K. Chu, *ACS Appl. Mater. Interfaces* **2016**, *8*, 13966–13972.
[31] B. Zheng, W. Zheng, Y. Jiang, S. Chen, D. Li, C. Ma, X. Wang, W. Huang, X. Zhang, H. Liu, F. Jiang, L. Li, X. Zhuang, X. Wang, A. Pan, *J. Am. Chem. Soc.* **2019**, *141*, 11754–11758.
[32] C. Lv, C. Yan, G. Chen, Y. Ding, J. Sun, Y. Zhou, G. Yu, *Angew. Chem. Int. Ed.* **2018**, *57*, 6073–6076.
[33] Y. Lin, C. Yang, S. Wu, X. Li, Y. Chen, W. L. Yang, *Adv. Funct. Mater.* **2020**, *30*, 2002918.
[34] Z. Zhu, H. Huang, L. Liu, F. Chen, N. Tian, Y. Zhang, H. Yu, *Angew. Chem. Int. Ed.* **2022**, *61*, e202203519.
[35] X. Zhao, X. Li, L. An, L. Zheng, J. Yang, D. Wang, *Angew. Chem. Int. Ed.* **2022**, *61*, e202206588.
[36] Q. Meng, C. Lv, J. Sun, W. Hong, W. Xing, L. Qiang, G. Chen, X. Jin, *Appl. Catal. B* **2019**, *256*, 117781.
[37] Y. Guo, W. Shi, Y. Zhu, Y. Xu, F. Cui, *Appl. Catal. B* **2020**, *262*, 118262.
[38] J. Yang, J. Jing, W. Li, Y. Zhu, *Adv. Sci.* **2022**, *9*, 2201134.
[39] Y. Gao, J. Zhu, H. An, P. Yan, B. Huang, R. Chen, F. Fan, C. Li, *J. Phys. Chem. Lett.* **2017**, *8*, 1419–1423.
[40] Z. Xing, J. Hu, M. Ma, H. Lin, Y. An, Z. Liu, Y. Zhang, J. Li, S. Yang, *J. Am. Chem. Soc.* **2019**, *141*, 19715–19727.
[41] Y. Wan, H. Zhou, M. Zheng, Z. H. Huang, F. Kang, J. Li, R. Lv, *Adv. Funct. Mater.* **2021**, *31*, 2100300.
[42] Y. Fang, Y. Xue, Y. Li, H. Yu, L. Hui, Y. Liu, C. Xing, C. Zhang, D. Zhang, Z. Wang, X. Chen, Y. Gao, B. Huang, Y. Li, *Angew. Chem. Int. Ed.* **2020**, *59*, 13021–13027.
[43] J. K. Nørskov, F. Abild-Pedersen, F. Studt, T. Bligaard, *Proc. Natl. Acad. Sci. USA* **2011**, *108*, 937–943.
[44] C. Yang, B. Huang, S. Bai, Y. Feng, Q. Shao, X. Huang, *Adv. Mater.* **2020**, *32*, 2001267.
[45] N. Zhang, L. Li, J. Wang, Z. Hu, Q. Shao, X. Xiao, X. Huang, *Angew. Chem. Int. Ed.* **2020**, *59*, 8066–8071.
[46] S. Liu, T. Qian, M. Wang, H. Ji, X. Shen, C. Wang, C. Yan, *Nat. Catal.* **2021**, *4*, 322–331.
[47] K. Chu, Q.-Q. Li, Y.-P. Liu, J. Wang, Y.-H. Cheng, *Appl. Catal. B* **2020**, *267*, 118693.

Manuscript received: March 15, 2023
Accepted manuscript online: May 25, 2023
Version of record online: June 13, 2023